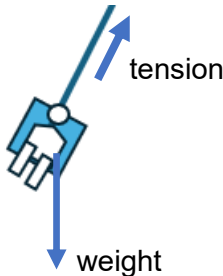


1	(a)	(i)	Because the speed at B is the same at C, the gain in KE is due to the loss of GPE from A to B only. Gain in KE in the end = Loss in GPE from A to B $\frac{1}{2}mv^2 = mgh$ $v = \sqrt{2gh} = \sqrt{2(9.81)(2.5)}$ $= 7.0 \text{ m s}^{-1}$	C1 A1
		(ii)	From A to B, the <u>gravitational potential energy drop is converted to kinetic energy</u> of the child. From B to C, <u>the further loss in gravitational potential energy is not converted into kinetic energy but dissipated as work done against contact friction</u> (or thermal energy and sound). Thus, the child's speed is unchanged.	B1 B1
		(iii)	Work done against friction = Fs $Fs = \text{loss of KE from C to D}$ $= KE_C - KE_D$ $F(7.6) = \frac{1}{2}(54)(7.0)^2 - 0$ $F = 168 \text{ N (3sf)}$ OR $v^2 = u^2 + 2a_{\text{ave}}s$ $0 = 7.0^2 + 2a_{\text{ave}}(7.6)$ average accel., $a_{\text{ave}} = -3.22 \text{ m s}^{-2}$ Average frictional force = $ma_{\text{ave}} = (52)(3.22)$ $= 168 \text{ N}$	M1 A1 (M1) (A1)
	(b)	(i)	When the carousel rotates, the chair and its occupant will travel in a straight-line if not for the cable attached to it. <u>A horizontal (centripetal) force is required</u> in order to pull the man towards the centre. This is provided by the <u>horizontal components of the tension</u> on the cable holding the chair. Thus, the cable has to tilt to provide for the needed centripetal force.	M1 A1
		(ii)	 All forces must be indicated and labelled in words (not in letters or symbols). <u>Tension</u> on cable and <u>gravitational force</u> (or <u>weight</u>) on person and seat. Deduct mark if FBD includes <i>centripetal</i> or <i>centrifugal</i> force, <i>normal</i> force or any irrelevant forces.	B1

2

		(iii)	Along the y-direction: $T \cos \theta = mg$ (balanced) Along the x-direction: $T \sin \theta = mr\omega^2$ (unbalanced due to centripetal rotation) $r = 4.7 + 8.4 \sin 35^\circ = 9.518 \text{ m}$ $\tan \theta = \frac{mr\omega^2}{mg} = \frac{r\omega^2}{g}$ $\tan 35^\circ = \frac{(9.518)\omega^2}{9.81}$ $\omega = 0.85 \text{ rad s}^{-1}$	C1 C1 A1
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2	(a)		The first law of thermodynamics states that the increase in internal energy of a	B1
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